Homework 5 Template: Model-Based and Offline RL

CMU 10-703: Deep Reinforcement Learning (Fall 2026)

Use this template to record your answers for Homework 5. Add your answers using LaTeX and then save your document as a PDF to upload to Gradescope. You are required to use this template to submit your answers. **You should not alter this template in any way** other than to insert your solutions. You must submit all 13 pages of this template to Gradescope. Do not remove the instructions page(s). Altering this template or including your solutions outside of the provided boxes can result in your assignment being graded incorrectly.

You should also zip your `code/` directory and upload it to the **separate** Gradescope coding assignment. Remember to mark all teammates on **both** assignment uploads through Gradescope.

Instructions for Specific Problem Types

On this homework, you must fill in blanks for each problem. Please make sure your final answer is fully included in the given space. **Do not change the size of the box provided.** For short answer questions you should **not** include your work in your solution. Only provide an explanation or proof if specifically asked. To include a plot, use `\includegraphics[width=\linewidth]{plot.png}` inside the box, scaled so that it fits.

Fill in the blank: What is the course number? 10703

Problem 0: Collaborators

Enter your team members' names and Andrew IDs in the boxes below. If you worked in a team with fewer than three people, leave the extra boxes blank.

Name 1:		Andrew ID 1:	
Name 2:		Andrew ID 2:	
Name 3:		Andrew ID 3:	

Collaboration: Detail the work division amongst your group below.

Solution

Kushal did Problem 1, Arnav did Problem 2 and Vineeth did Problem 3

Part I: Model-Based RL

Problem 1: Model-Based RL with PETS (105 pts)

1.1 Model-Based Predictive Control (25 pts)

1.1.1 (10 pts) CEM (without MPC) with ground-truth dynamics.

Success percentage: 94 %

1.1.2 (10 pts) Random sampling with ground-truth dynamics.

Success percentage without MPC: 70 %

Success percentage with MPC: 94 %

How does the performance of random sampling compare to that of CEM?

Solution

Both methods sample actions from the same initial Gaussian distribution, but CEM iteratively updates its sampling distribution using elite trajectories, while random shooting samples all trajectories independently and only selects the best at the end. Thus CEM generally finds better open-loop plans. MPC improves random shooting by replanning after every executed action, correcting errors caused by imperfect plans which explains its higher success rate.

1.1.3 (5 pts) MPC vs. open-loop control: which performs better on what environments, and why? Discuss the pros and cons of MPC.

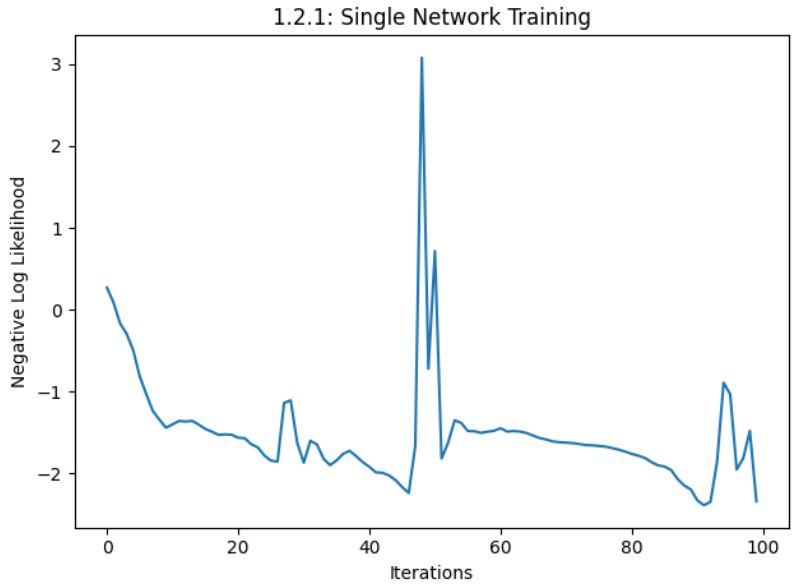
Solution

For Pushing2d-v1, we get 86% and 96 % without and with MPC respectively. For Pushing2dNoisyControl-v1, we get 58 % and 80 % without and with MPC respectively.
MPC is expected to perform better in environments with noisy actions/stochastic dynamics becuase it repeatedly observes the true state and replans.
Open-loop control can be preferable in deterministic environments with accurate dynamics where a planned sequence is already reliable. MPC's main advantage is replanning after every executed action, which allows is to correct errors. MPC's main disadvantages are its computational cost and with a long horizon or expensive optimizer, replanning can be slow.

1.2 Single Probabilistic Network (30 pts)

1.2.1 (4 pts) Training loss plot.

Solution



1.2.2 (10 pts) MPC with random sampling. Success percentage: 18 %

1.2.3 (10 pts) MPC with CEM. Success percentage: 9 %

1.2.4 (6 pts) Random sampling vs. CEM. How did MPC with this model perform in general? When does the derived policy succeed, and when does it fail?

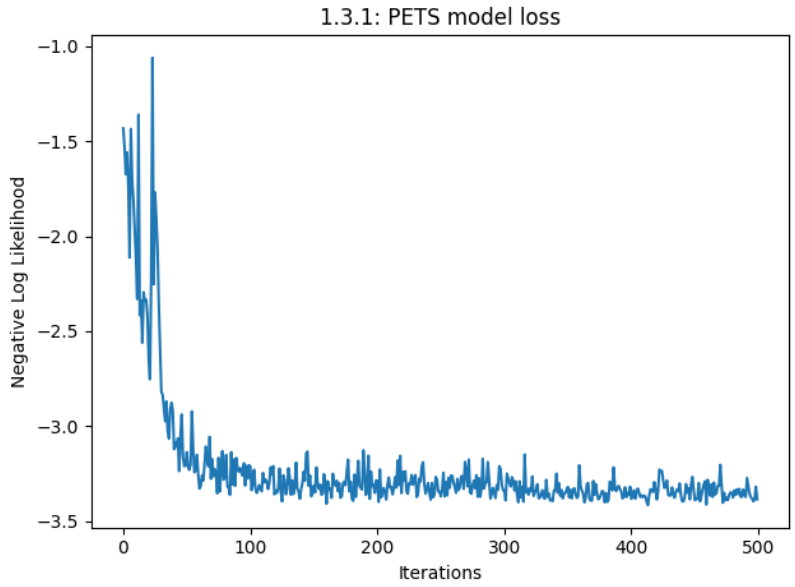
Solution

Random shooting achieved an 18% success rate, while CEM achieved 6%. Although CEM is a more effective optimizer in the ground-truth setting, it optimized inaccuracies in the learned dynamics model. This caused it to select action sequences with low predicted cost but poor real-environment performance. Random shooting was less susceptible to this model exploitation and therefore **performed better in this experiment**.
The learned-model policy succeeded when the model predictions were accurate over the short planning horizon and the required motion was simple. It failed especially near contact dynamics, in states poorly represented in the random training data, and when prediction errors compounded across several imagined steps. MPC helped by replanning, but it could not fully correct systematic errors in the learned model.

1.3 MBRL with PETS (15 pts)

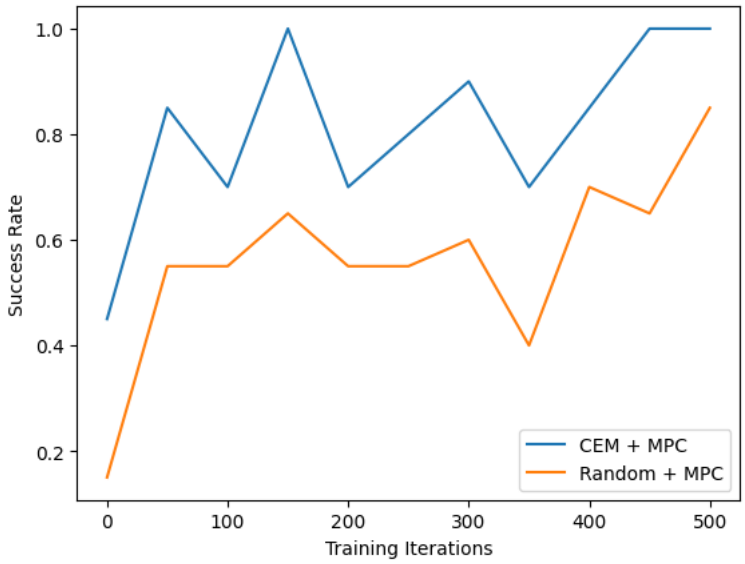
1.3.1 (10 pts) Training loss plot.

Solution



1.3.2 (5 pts) Test success percentage vs. PETS iterations plot.

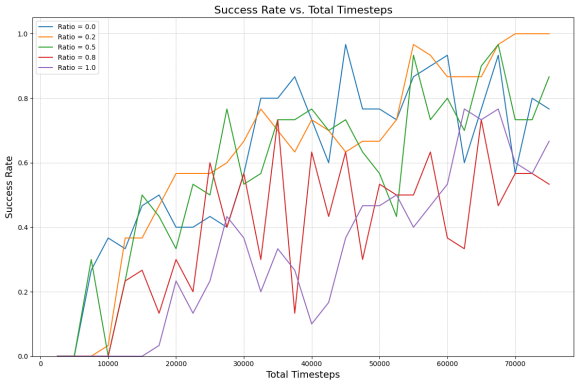
Solution



1.4 Training with Synthetic Data (35 pts)

1.4.1 (10 pts) Ablation over model vs. real data: plot and discussion.

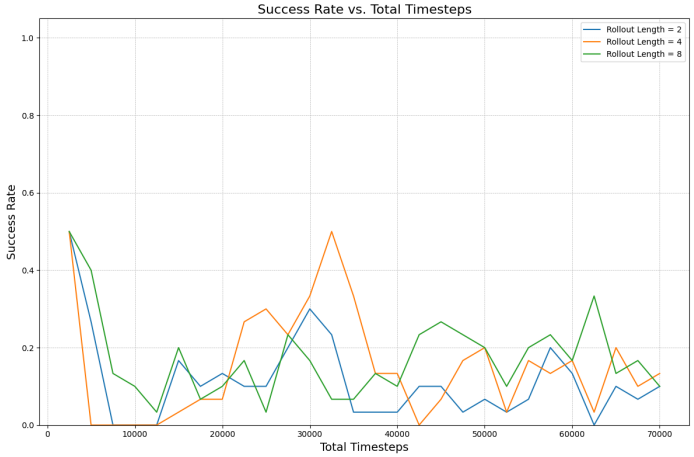
Solution



The results are noisy and do not show a clear monotonic relationship between the synthetic-data ratio and success rate. In general we observe runs with low amount of synthetic data perform reasonably well agsainst runs with high amounts of synthetic data. Overall, these results suggest that a small or moderate amount of synthetic data can be useful, but relying entirely on model-generated data can lead to unstable performance. Real data anchors TD3 to actual environment behavior, while synthetic data increases the amount of experience available for training.

1.4.2 (10 pts) Ablation over model rollout lengths: plot and discussion.

Solution



The success-rate curves are highly variable and show no clear trend, and all of them result in poor success rates. In general we expect shorter rollouts to perform better because they limit compounding errors. Longer rollouts provide more synthetic transitions per starting state, but they also make TD3 more dependent on model predictions farther from the real data distribution.

1.4.3 (10 pts) Model error vs. rollout length: plot and discussion.

Solution



Unlike the success-rate plots, the model-error plot shows a clear pattern. The mean L2 prediction error increases steadily with rollout step, from approximately 0.15 at the first step to approximately 0.5 by step 15. This confirms that model errors compound as predictions are fed back into the model. The shaded area also increases overall, indicating greater variance at longer rollouts which is also a consequence of compounding errors

1.4.4 (5 pts) Under which scenarios would you prefer MBRL to policy gradient methods?

Solution

MBRL is preffered when environment interaction is expensive or inefficient, because a learned dynamics model can generate many imagined or hypothetical transitions from each real interaction. MBRL can also be useful when the system has reasonably predictable/determinisitic dynamics and planning can exploit that structure. Policy-gradient methods may be preferable when the dynamics are highly complex or difficult to model accurately. They can learn directly from real experience without suffering from long horizon errors (as seen in 1.4.2).

Part II: Offline RL

Problem 2: TD3 + Behavior Regularization (50 pts)

2.2 Running TD3 without Any Offline Learning Modifications

Learning curves (evaluation returns and Q-values), and: what insights can we draw from evaluation returns being low but predicted Q-values being high?

Solution

2.3 Behavior Regularized TD3

Learning curves (evaluation returns and Q-values) with $\alpha_{\text{reg}} = 10$.

Solution

2.4 Conceptual Questions about Picking α_{reg}

1. For which kinds of datasets will we expect larger values of α_{reg} to work better? For which datasets will we expect lower values to work better?

   Solution

2. Given a completed run with $\alpha_{\text{reg}} = 1.0$, how would you choose a reasonable value of α_{reg} if you could only run a single more run?

Solution

Problem 3: Conceptual Questions on Offline RL (50 pts)

(i) Select all that apply and justify (see the handout for the three methods and statements 1–3).

Solution

(ii) Describe a scenario where distribution constraint methods would be preferable to support constraints.

Solution

(iii) Select all that apply and justify (see the handout for statements 1–3).

Solution

(iv) Why can offline RL perform better than imitation learning? Give a dataset where offline RL should outperform imitation, and one where imitation should outperform offline RL.

Solution

Feedback

Feedback: You can help the course staff improve the course for future semesters by providing feedback. What was the most confusing part of this homework, and what would have made it less confusing?

Solution

Time Spent: How many hours did you spend working on this assignment? Your answer will not affect your grade. Please average your answer over all the members of your team.

Solution

Alone	
With teammates	
With other classmates	
At office hours	